



Forecasting Polish-Airport Seat Supply

from Origin-Market Macroeconomic Inputs

A one-equation forecaster for IMF-scenario translation into airline capacity

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Executive summary

One formula, three usable outputs, a single macro input

① What it is

A single-elasticity forecaster for monthly seats offered at Polish airports, driven by aggregate origin-market GDP at the airline planning lag.

② What it returns

Monthly seats(t) for any horizon up to 12 months ahead. Reads directly from IMF / WEO macro projections.

③ Where it adds value

Scenario differentiation: a 5-pp spread in projected GDP growth translates into a 0.5M-seat spread in the forecast (~9% of average monthly capacity).

④ What it is not

Not a demand model (overnight stays are in NB05–NB08). Not a black box: the formula is a single line with two coefficients.

The question

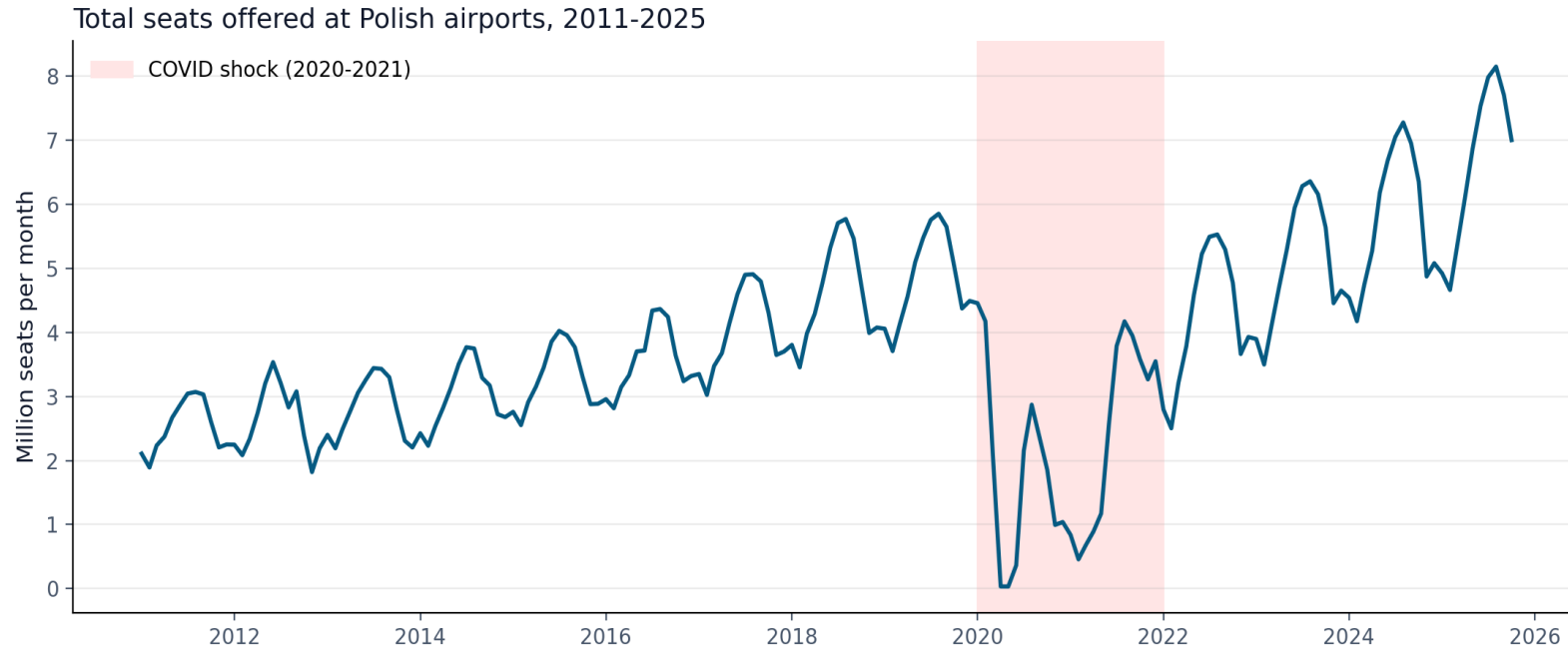
- Can macroeconomic conditions in origin markets be turned into a forecast of seats offered at Polish airports?
- With a formula a planner can read?



Two parallel models are estimated: a linear OLS that produces the closed-form formula, and an XGBoost benchmark that tests whether non-linearity or additional macro channels would add precision.

The target — seats offered at Polish airports, 2011-2025

Eurostat avia_tf_aca, summed across 13 Polish commercial airports — monthly



THREE OBSERVATIONS

1 Structural growth

Total seats nearly quadrupled in 14 years — supply-side capacity expansion (LCC base openings, fleet growth), not macro-driven.

2 Sharp COVID shock

April 2020 reached 1.6% of January 2020 levels. Recovery in place by summer 2022.

3 Annual seasonality

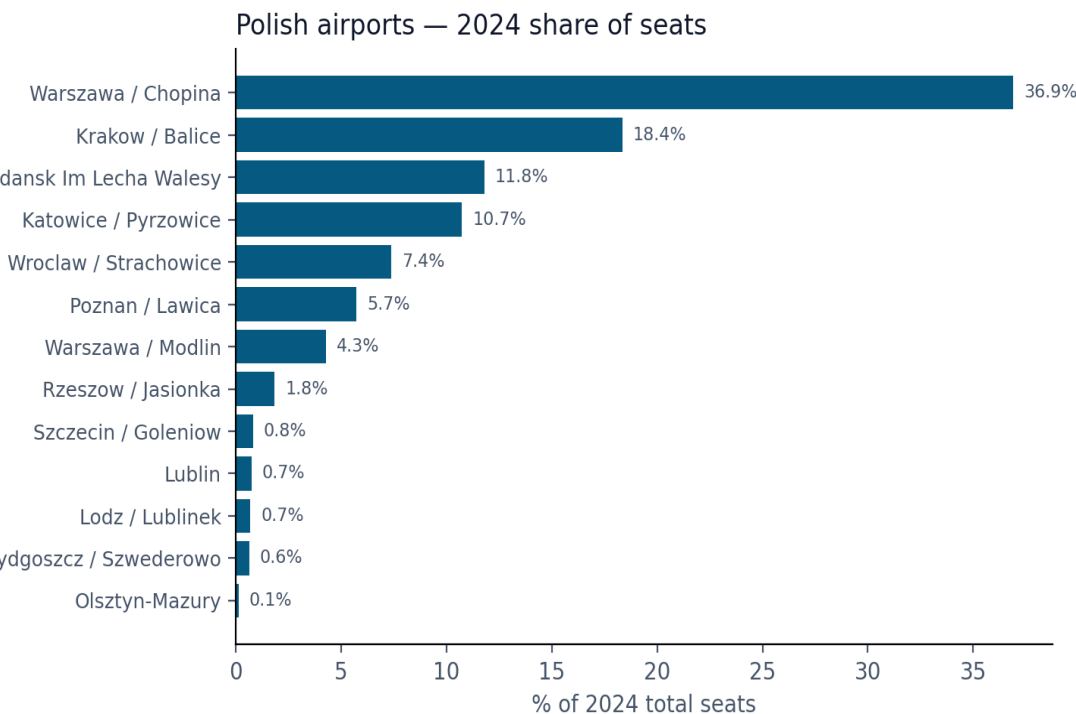
Clear summer peak vs winter trough, preserved across the COVID dislocation.

Why this shapes the model

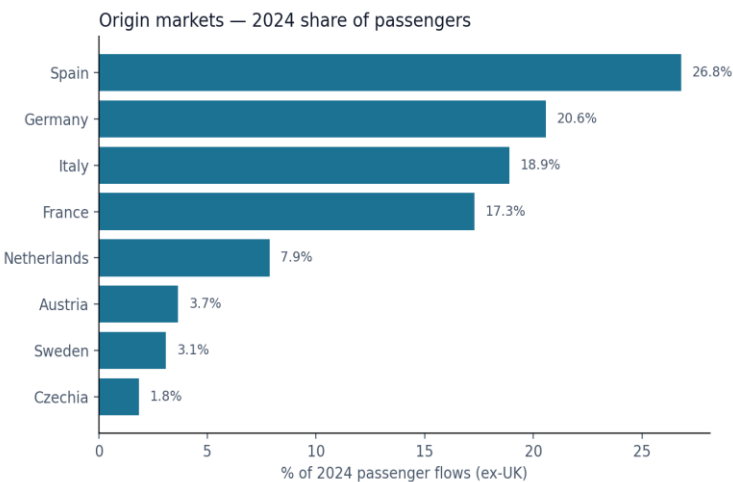
Most seat growth is structural and supply-side. Origin-market macro variables vary by ± 1 -2 pp per year while seats vary by ± 10 pp. Any macro signal must be teased out of the residual cyclical variation around the trend.

Composition of the total

Two angles on the same aggregate — capacity by Polish airport, passenger flows by origin

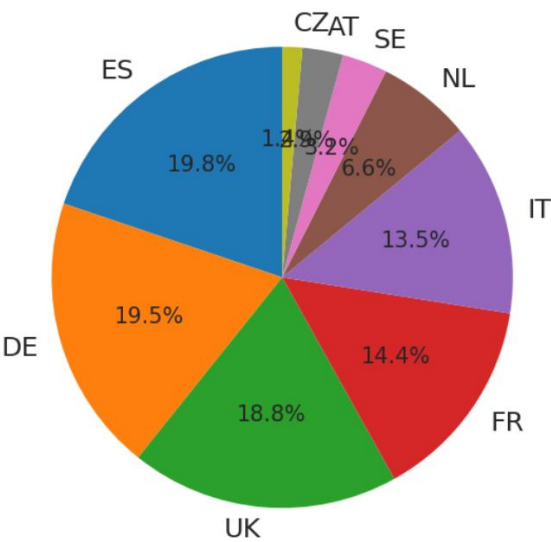


Concentration: Warsaw Chopin alone holds ~37% of 2024 seat capacity; the top four airports cover ~78%. Forecasting the national total effectively means tracking five hubs.



Origin mix: Spain, Germany, Italy and France together account for ~84% of 2024 passenger flows (ex-UK). UK data unavailable in Eurostat post-Brexit; the panel uses ONS chain-linked.

Market share by origin country



Approximation: Considering the historical data (to include UK), the distribution spreads including UK as the **third major origin market** at almost 19% of the total. This shows that Spain has grown even more in the last few years, with UK being also one of the main relevant countries.

The modelling approach

Three deliberate methodological choices, each motivated by what the data tells us

01

Year-on-year log differences

We compare each month against the same month one year earlier. This removes both the secular growth trend and the annual seasonal cycle by construction, leaving stationary residual variation to explain.

02

Macro regressor at the 12-month lag

Airline capacity for month t is committed roughly 12 months earlier — IATA slot conferences, schedule publishing and ticket-selling windows all run on an annual cycle. Contemporaneous macro is therefore information that arrives too late to drive seats.

03

Aggregate origin-market GDP

Seats are a raw count, so the driver should also be aggregate: $\text{gdp_total} = \text{GDP per capita} \times \text{population}$. Per-capita GDP gives a near-identical statistical fit but a less natural reading for an aggregate target.

The forecasting formula

Plug-in form, ready for a spreadsheet

$$\text{seats}(t) = \text{seats}(t-12) \times \exp(0.074 + 1.57 \cdot \text{dlog_gdp_total}(t-12))$$

$$\text{dlog_gdp_total}(t-12) = \ln(\text{gdp_total}(t-12)) - \ln(\text{gdp_total}(t-24))$$

$$\text{gdp_total}(t) = \text{gdp_pc_origins}(t) \times \text{population_origins}(t)$$

STRUCTURAL GROWTH (A)

+7.4%

per year, independent of macro

INCOME ELASTICITY (B)

1.57

IATA canonical range 1.5-2.5

MACRO INPUT FREQUENCY

Annual

GDP YoY change, lagged 12 months

Reading the coefficients

Each coefficient maps onto an institutional behaviour an economist can verify against literature

$$\alpha = +0.074$$

STRUCTURAL GROWTH OF SEATS

≈ 7.4% per year

of seat growth that the macro channel does not explain.

What it captures

- LCC base openings (Wizz Air, Ryanair)
- Fleet expansion and replacement
- Strategic route launches
- Long-run capacity decisions that do not respond proportionally to year-to-year macro fluctuations

$$\beta = +1.57$$

INCOME ELASTICITY (TO ORIGIN GDP)

1 pp ↑ in YoY origin GDP →

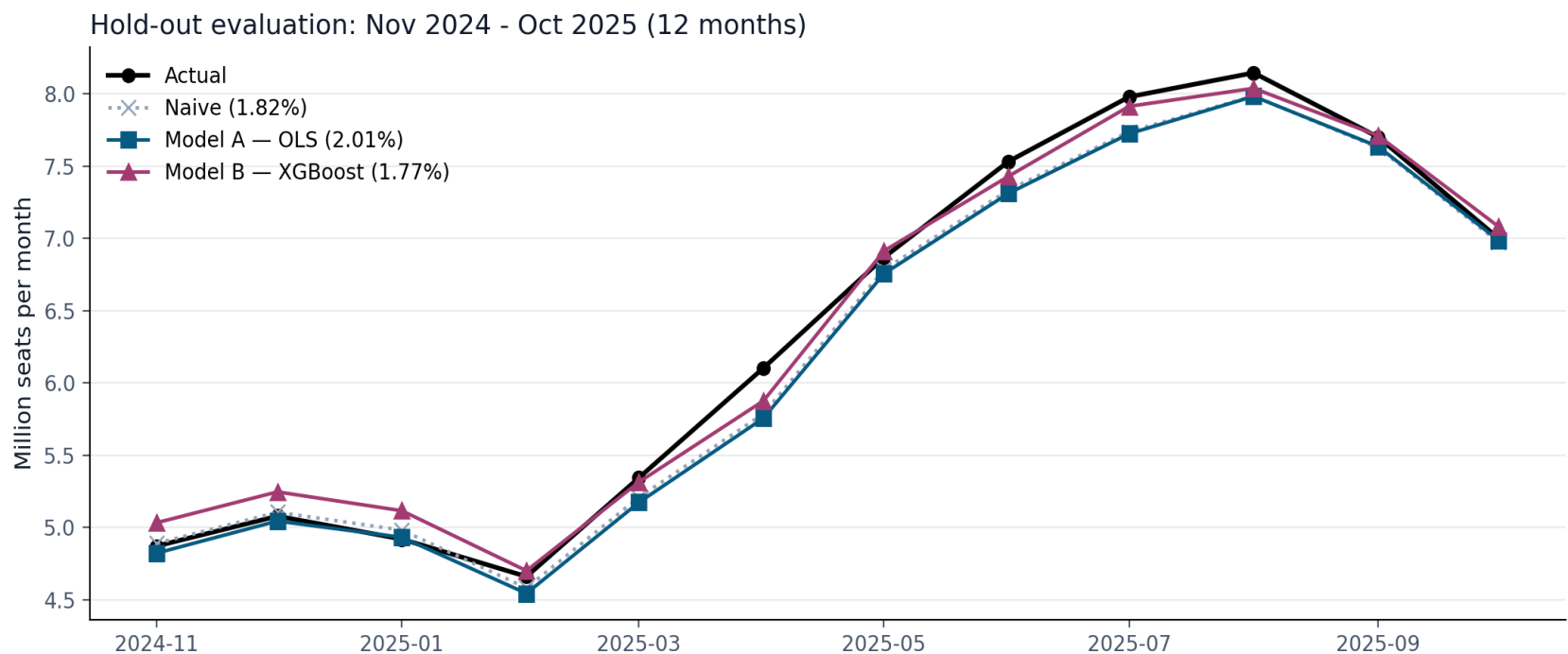
1.57 pp ↑ in YoY seats growth, twelve months later.

How it compares to literature

- IATA / Boeing report income elasticities in the 1.5-2.5 range for European short-haul aviation
- Our estimate sits exactly at the lower end of that range
- Marginally significant ($p = 0.050$) — wide confidence interval, but the point estimate is economically defensible

Hold-out accuracy

12 months held out (Nov 2024 - Oct 2025), same window for all three forecasters



Hold-out MAPE

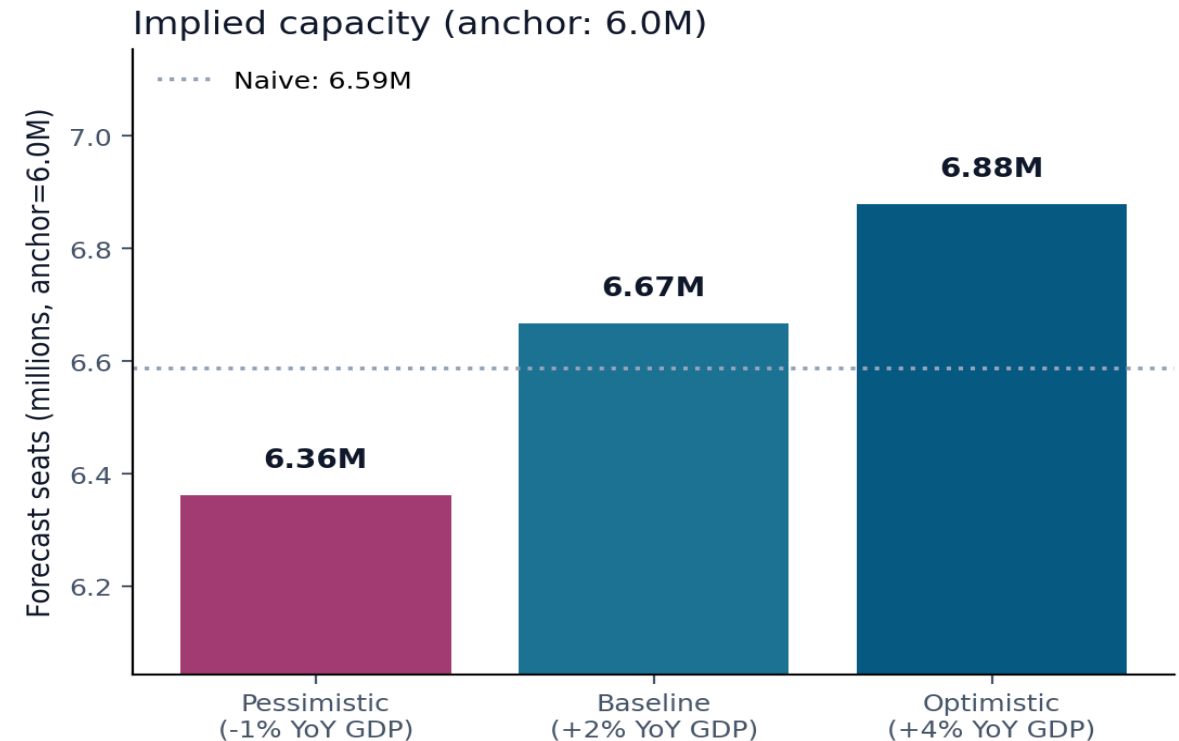
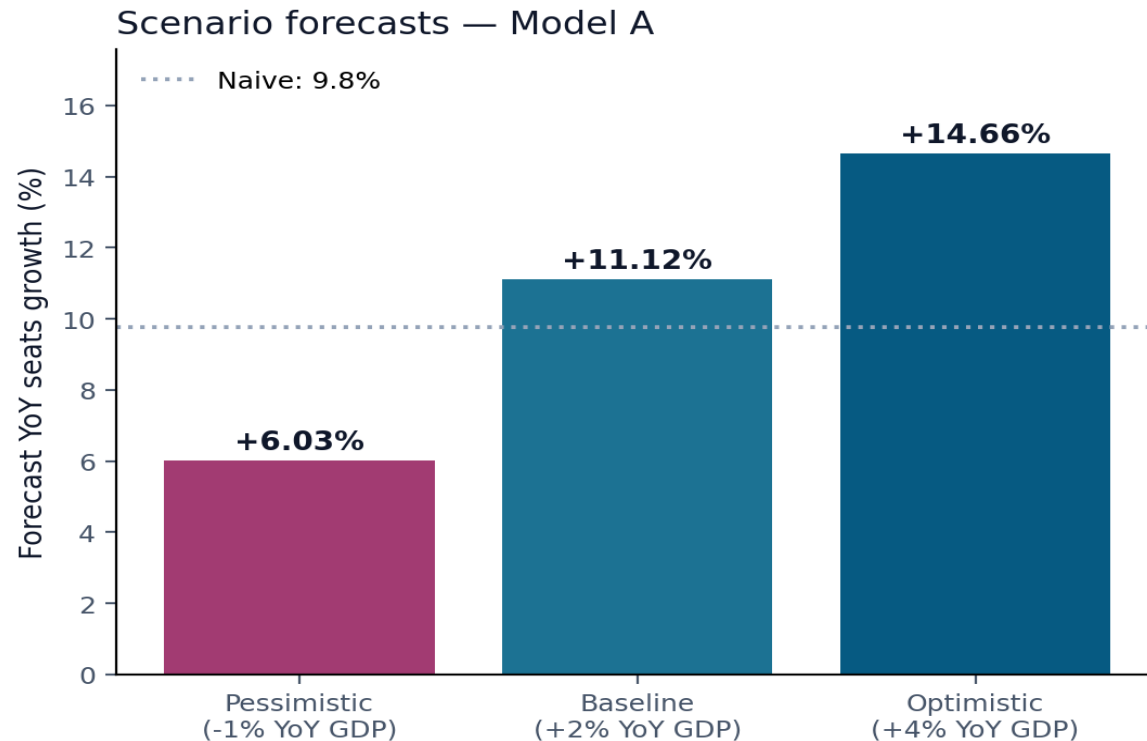
Naive <i>seats(t-12) × 1.098</i>	1.82%
OLS (the formula) <i>1 macro feature</i>	2.01%
XGBoost <i>7 lagged features</i>	1.77%

The honest reading

All three approaches cluster within a quarter of a percentage point of each other. At a 12-month horizon, the seasonal anchor $\text{seats}(t-12)$ does most of the predictive work. XGBoost extracts ~ 0.05 pp of additional signal over naive — small but real, confirming that lagged macro carries information. The OLS gives up ~ 0.2 pp of MAPE relative to XGBoost in exchange for one transparent equation.

Where the model adds value: scenario analysis

Three IMF / WEO paths for origin-market aggregate GDP, plugged directly into the formula



What the spread looks like

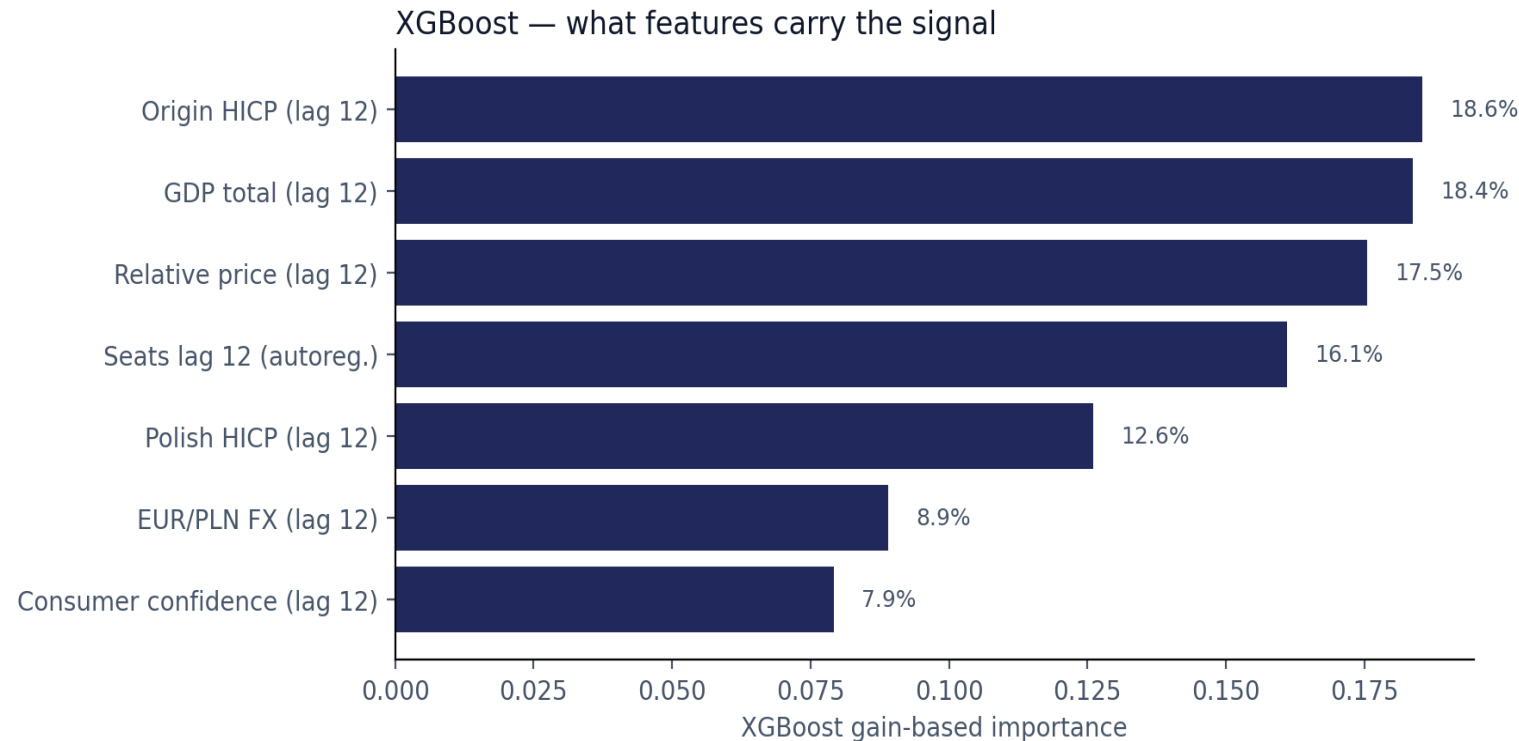
A 5-pp swing in projected origin GDP growth (-1% to +4%) translates into a 0.52M-seat swing in the forecast — roughly 9% of average monthly capacity.

Why naive can't do this

The naive forecast gives the same number regardless of macro scenario (+9.78% YoY). The formula reads the macro lever directly — and that is its operational purpose.

XGBoost — where does the additional signal come from?

Importance distributed across seven lagged macro features — GDP is not alone



WHAT THIS TELLS US

No dominant feature

The top four features each carry 16-19% importance. The signal is spread, not concentrated.

Inflation matters too

Origin HICP and relative price together carry as much weight as GDP. Real income, not nominal, drives capacity.

The autoregressive term sits mid-table

Seats lag 12 (~16%) confirms the seasonal anchor matters but does not dominate the macro channels.

Implication for the OLS

A multi-regressor OLS with relative price and consumer confidence at lag 12 was tested. In-sample R^2 rises (0.33), but hold-out MAPE degrades (2.16%) — **classic small-sample overfit**. The single-elasticity OLS gives up some statistical bells and whistles in exchange for a parsimony that generalises.

Conclusions and recommended use

Four takeaways for translating IMF projections into Polish-airport capacity forecasts

01 Use the formula

One equation, one elasticity ($\beta = 1.57$, in IATA range), one input. Implementable in a spreadsheet.

02 Don't expect to beat the naive at 12 months

All three forecasters cluster within 0.25 pp MAPE. The **structural growth and seasonal anchor** dominate at this horizon.

Still, more data needs to be collected (currently impossible to predict the trend due to COVID-19 crash).

03 Use it for scenario differentiation

A 5 pp swing in IMF GDP path $\rightarrow$ 0.5M-seat swing in forecast (~9% of capacity). Naive can't do this.

04 Use it for horizons > 12 months

When the naive seasonal anchor expires, the macro elasticity gives a coherent way to roll forecasts forward year by year.